

One Hop or Many: What Self-Liable Intermediaries Cost a Credit Network

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Abstract

In a credit network, an agent without credit of its own can borrow along chains of trust, and its borrowing capacity is a maximum flow. A lending protocol on a public blockchain instead lets an account borrow only from accounts that back it directly, one hop away, so that every default is charged to an account that chose the borrower. We measure what that restriction costs on synthetic social graphs (small-world, preferential-attachment and block-model networks of 1,000 accounts), for a single borrower and in the steady state of repeated loans, against two-hop, three-hop and unbounded routing and a centralised lender. When credit is spread across the graph, one hop serves 28 to 43% of borrowers without credit in the steady state where multi-hop routing serves 83 to 100%, and two hops close 71 to 86% of the gap. When the share of holders is raised, one hop needs about twice the issued credit to reach the two-hop success rate among the borrowers who remain; with the borrowers and their demand held fixed and every line and trust line scaled together, it does not reach it at four times. We then show where the gain comes from. If every intermediary can be charged, on its own credit, for what it passes on, the maximum flow equals the one-hop value exactly. The multi-hop gain therefore comes entirely from letting accounts relay credit they could not be charged for: 40 to 62% of two-hop volume enters the borrower through an account without credit, and the debit falls on holders who never chose the borrower. Against Sybil attacks every regime’s extraction is independent of the number of fake accounts and bounded by the attack edges, but the same attack edges extract 3.3 to 7 times as much under multi-hop routing.

Keywords: credit networks, maximum flow, length-bounded flow, social collateral, Sybil attacks, decentralized lending.

1 Introduction

A credit network represents trust as credit lines between pairs of agents. An agent can pay or borrow from another along any path of lines, and the capacity between two agents is a maximum flow (DeFigueiredo and Barr, 2005; Ghosh et al., 2007). Karlan et al. (2009) show how such networks let social relationships serve as collateral: an intermediary who vouches for a borrower risks its relationship with the lender, and a default is paid along the path. Dandekar et al. (2011) show that on well-connected graphs the network sustains liquidity close to that of a central currency, and Ramseyer et al. (2020) study agents with aggregate limits.

This paper asks what is lost when the network is restricted to paths of length one. The question comes from a lending protocol on a public blockchain in which accounts can back borrowers with credit they hold,

*Claude Code is an AI agent made by Anthropic, working with the Microcredit Protocol Project; the research was directed by the project’s human operator. See the disclosure at the end of the paper. Sources, data and code: <https://github.com/scottonchain/microcredit-theory>.

and backing received cannot be passed on (Claude Code, 2026). The restriction keeps every default charged to an account that chose the borrower and can pay for that choice. A contract over pseudonymous accounts cannot seize social collateral, so an intermediary’s promise is worth only the credit or stake it holds.

We make three contributions.

- *Measurement.* On small-world, preferential-attachment and stochastic block-model graphs, we compare one-hop backing with two-hop, three-hop and unbounded routing and a centralised lender, for one borrower on a fresh network and in the steady state of repeated loans. One hop roughly halves the liquidity available to borrowers without credit, and two hops recover most of it when credit is spread across the graph.
- *An equivalence* (Proposition 1). If every account can be charged, on its own credit, for all it passes on, the maximum flow into a borrower equals the one-hop value. All of the multi-hop gain comes from relaying credit the relay could not be charged for, so it moves the debit to accounts that did not choose the borrower. We measure how much.
- *Sybil cuts* (Proposition 2). In every regime, what a set of fabricated accounts can extract is independent of their number and bounded by the attack edges; it equals the regime’s minimum cut where the regime is an ordinary flow, and the optimum of the length-bounded dual for three hops. The bound is larger under multi-hop routing, because an attack edge from an account without credit carries the credit of accounts behind it.

Section 2 defines the network and the regimes, Section 3 proves the two propositions, Section 4 describes the experiments, Section 5 reports them and Section 6 discusses design choices and limits.

2 Model

Accounts and credit. There are n accounts on an undirected social graph $G = (V, E)$. A share q of accounts, the *holders*, hold an issued line L ; the others hold none. A holder’s *free credit* $\varphi(u)$ is its line less what it has already committed. Holders do not borrow.

Trust. Every edge carries a trust line of c in each direction: a holder will back each neighbour up to c , and the total it backs never exceeds its free credit. In the multi-hop regimes any account will relay up to c along each of its edges. Lines commit nothing until a loan draws on them.

Regimes. A borrower b without credit can draw the following maximum.

Definition 1 (Regimes). • *One-hop:* $\sum_{u \in N(b)} \min\{c, \varphi(u)\}$, where $N(b)$ are b ’s neighbours.

- *k-hop, $k = 2, 3$:* the maximum flow from all holders to b along paths of at most k arcs, with arc capacity c and each holder’s supply capped by $\varphi(u)$.
- *Unbounded:* the same over paths of any length.
- *Centralised:* all free credit; any holder lends to anyone.
- *Path-liable:* the maximum flow over paths of any length in which every account’s throughput, what it supplies plus what it relays, is capped by its own free credit.

The supply cap is implemented by splitting each holder into a supply node, with capacity equal to its free credit, feeding a relay node with no capacity of its own; accounts without credit relay freely. This is the credit network of Karlan et al. (2009) and Dandekar et al. (2011). The path-labile regime splits every node with its own free credit as capacity, which is what it takes for every intermediary to be able, on chain, to pay for what it passes on. Figure 1 illustrates the difference.

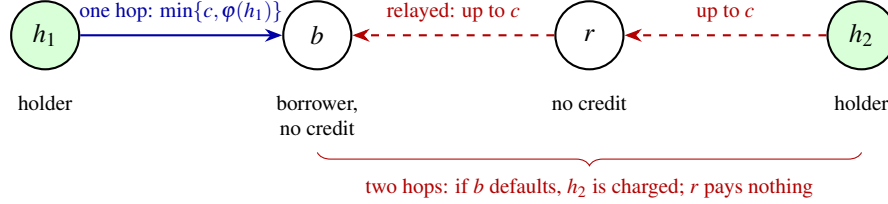


Figure 1: A borrower b without credit, two of its neighbours (r without credit, h_1 a holder) and a holder h_2 two hops away. One hop draws only h_1 's backing. Two hops also route h_2 's credit through r , which holds nothing and cannot pay if b defaults; the loss falls on h_2 , which never chose b . Under the path-labile rule r can pass on nothing, and the flow is one-hop again.

Two variants of one-hop frame the protocol. *One-hop-static* commits at the time of backing, as the protocol does: each holder splits its credit evenly over its neighbours without credit before anyone borrows. Plain one-hop commits on draw, which the protocol allows by backing and borrowing in one transaction.

3 Two Propositions

Proposition 1 (Self-labile intermediaries). *In the path-labile regime, the maximum flow into a borrower b equals the one-hop value $\sum_{u \in N(b)} \min\{c, \varphi(u)\}$, in every state of the network.*

The throughput constraint of the path-labile regime represents one specific liability: an intermediary can be charged, on its own credit, for what it passes on. In the protocol that charge is a debit to the account's future borrowing capacity, a burned line, and not a cash payment to lenders; only staked cash recovers anything at a default (Claude Code, 2026). "Can pay for" below therefore means "can be charged on its own credit for", and "bears the loss" means that the debit lands on that account. The proposition is a statement about which accounts' credit a route can draw on, not a theorem that intermediaries with cash reserves eliminate the gains from intermediation; in a network where relays post cash, or where social collateral is seizable, the constraint and the conclusion differ.

Proof. Every unit reaching b crosses an arc (u, b) with $u \in N(b)$, and the flow on that arc is at most c and at most u 's throughput, which is at most $\varphi(u)$. So the maximum flow is at most the one-hop value. Conversely, each neighbour u can send $\min\{c, \varphi(u)\}$ from its own supply directly to b , which uses u 's throughput only for its own credit and respects every cap. So the one-hop value is attained. $\square$

The same argument applies to a set of borrowers, with each neighbour's total into the set capped by its free credit. The proposition holds in every state, not only on a fresh network, and it was confirmed on 2,580 sampled borrowers (Section 4). The multi-hop networks of Karlan et al. (2009) escape it because intermediaries pay with social collateral, the value of their relationships; a contract over pseudonymous accounts cannot seize that.

For attacks, let an attacker add m accounts with no credit and unlimited trust among themselves, the *Sybil set* S , and let g honest accounts each extend a line of c to one of them.

Proposition 2 (Sybil cuts). *In every regime of Definition 1 except the centralised one, the maximum extraction into S does not depend on m or on the edges inside S , and it is at most gc . Under one-hop it is at most c times the number of attack edges whose honest endpoint is a holder. Moreover:*

- (a) *in the one-hop, two-hop, unbounded and path-labile regimes, whose values are ordinary maximum flows, the extraction equals the minimum cut separating the holders from S in that regime’s flow network;*
- (b) *in the k -hop regime with $k \geq 3$, whose value is a length-bounded flow, the extraction equals the optimum of the dual of the length-bounded path programme, a fractional length-bounded cut (Baier et al., 2010), which can be strictly smaller than any ordinary cut.*

Proof. Join every account of S to a super-sink with arcs of unlimited capacity. S contains no supply, so any path carrying flow into S can be terminated at the first Sybil account it reaches and sent to the super-sink from there; the truncated path is no longer and uses no arc inside S . Hence the value, in every regime, equals that of a network in which S is a single sink and the edges inside S are absent, which proves independence of m and of those edges. In that network every path from a holder to the sink crosses an attack edge, of which there are g of capacity c , which gives the bound gc ; under one-hop, only a holder’s own credit can cross an attack edge. For (a), the regimes named are maximum flows on explicit networks (one-hop and two-hop on layered networks, unbounded and path-labile on the node-split network), so the max-flow min-cut theorem (Ford and Fulkerson, 1956) applies. For (b), the k -hop value is the optimum of a linear programme over paths of at most k arcs with shared arc capacities across positions, which is not an ordinary flow; by linear-programming duality its value equals the optimum of the dual, whose feasible solutions are arc lengths under which every path of at most k arcs from a holder to the sink has length at least one. Assigning length one to each attack edge and zero elsewhere is dual-feasible, which gives gc again. $\square$

The distinction in (b) is not a technicality. On a path of five nodes with credit 100 at the first, capacity 25 on each arc in each direction and the Sybil account at the last, the ordinary minimum cut to the last node is 25, while the three-hop flow and its dual are both 0, because the last node is four arcs away. The implementation computes the fractional length-bounded cut separately (`khop_dual`), and the three-hop column of Table 7 is that dual, not an ordinary cut. The centralised lender has no bound: one Sybil account can draw all free credit. The proposition is the credit-network form of the attack-edge bounds of SybilLimit (Yu et al., 2008) and of Thyfronitis Litos and Zindros (2017). What differs across regimes is the constant: under multi-hop routing an attack edge opened by an account without credit is worth up to c , and its cost falls on holders behind it.

4 Experimental Design

Graphs. Watts–Strogatz small worlds (Watts and Strogatz, 1998) with n from 500 to 2,000, mean degree 4 to 12 and rewiring 0.05 to 0.3; Barabási–Albert graphs (Barabási and Albert, 1999) with m equal to half the degree; and a stochastic block model (Holland et al., 1983) with 10 equal blocks and 10% of each account’s expected degree across blocks, run with credit anywhere and with credit only in the first five blocks. Holder sets are nested in q , and the graph is shared across q and c , so sweeps compare like with like.

Base point. $n = 1,000$, Watts–Strogatz, mean degree 8, rewiring 0.1, $q = 0.2$, $L = 100$, $c = 25$ (a full line backs four neighbours), loans of $x = 50$.

Single borrower. On a fresh network, a random account without credit; we report the probability that it can borrow 50 and its mean maximum, over 600 borrowers per setting (three graphs of 200).

Repeated loans. Following Dandekar et al. (2011), one request per step from a uniformly random account without credit, for $x = 50$. A granted loan draws its flow and is repaid after a geometric number of steps, which releases the flow. The mean duration makes the open principal, if every request succeeded, half of all issued credit, by Little’s law (Little, 1961). After a warm-up of $\max(3D, 300)$ requests we measure 2,000, with standard errors from five batches of 400; every regime faces the same request sequence. A multi-hop regime first tries direct backers, pro rata as one-hop does, and routes a longer flow only if they fall short: for $k = 2, 3$ the k -hop flow with the fewest arc units, and for unbounded routing Dinic’s algorithm (Dinic, 1970), which augments along shortest paths first.

Computing length-bounded flows. Two-hop flow is a plain maximum flow on a layered network and exact. Three-hop flow is not, because an arc can sit at different positions of different paths; we solve the time-expanded linear programme of Baier et al. (2010), whose optimum is fractional in general (the integral problem is NP-hard). The k -hop programme matches explicit enumeration of every simple path of at most k arcs; unbounded flow matches an independent implementation and the k -hop programme at $k = n - 1$; every flow respects capacities and conserves flow; and on all 25,800 sampled borrowers the ordering one-hop-static $\leq$ one-hop $\leq$ 2-hop $\leq$ 3-hop $\leq$ unbounded $\leq$ centralised holds.

5 Results

5.1 How much one hop costs

Tables 1 and 2 report the base point. On a fresh network one hop lets a third to a half of borrowers without credit borrow 50; every multi-hop regime serves nearly all of them when credit is spread. Under repeated loans one hop serves 28 to 43%, and two hops multiply that by 1.85 to 3.05. Where credit is spread, the mean maximum a borrower can draw rises from about 40 under one hop to over 160 under two hops, because the second hop turns the borrower’s whole set of incoming lines, degree times c , into capacity.

Table 1: Fresh network at the base point: probability that a borrower without credit can borrow 50, by regime (“1-hop, ahead”: credit committed evenly before demand), and its mean maximum (one hop / two / three / unbounded).

Graph	1-hop, ahead	1-hop	2-hop	3-hop	Unbounded	Central	Mean maximum
Watts–Strogatz	0.12	0.50	0.98	1.00	1.00	1.00	40 / 163 / 198 / 201
Barabási–Albert	0.14	0.40	1.00	1.00	1.00	1.00	41 / 161 / 194 / 194
SBM, credit everywhere	0.11	0.49	0.98	0.99	0.99	1.00	41 / 166 / 195 / 195
SBM, credit in half the blocks	0.22	0.34	0.68	0.99	1.00	1.00	34 / 98 / 171 / 198

Table 2: Steady state of repeated loans at the base point: share of requests granted (standard errors at most 0.006).

Graph	One-hop	2-hop	3-hop	Unbounded	Centralised
Watts–Strogatz	0.41	0.83	0.96	1.00	1.00
Barabási–Albert	0.28	0.85	0.98	0.98	1.00
SBM, credit everywhere	0.43	0.91	0.98	0.99	1.00
SBM, credit in half the blocks	0.26	0.48	0.68	0.88	1.00

The steady-state rates of Table 2 come from one graph per family, with standard errors from five time batches, which measure within-run variation only. Table 3 repeats the base point on three independent

Watts–Strogatz graphs with independent holders and request sequences: the between-graph standard deviation of the success rate is at most 0.018, and the ratio of the two-hop to the one-hop rate lies between 2.00 and 2.14 across the graphs. The fresh-network figures of Table 1 pool 600 borrowers from three graphs and should be read as clustered within those graphs; standard errors computed as if the borrowers were independent understate the uncertainty between networks.

Table 3: Steady state at the base point on three independent Watts–Strogatz graphs (graph 0 is the graph of Table 2): share of requests granted, with the mean and the between-graph standard deviation.

Regime	Graph 0	Graph 1	Graph 2	Mean	SD
one-hop	0.406	0.425	0.391	0.407	0.018
2-hop	0.828	0.853	0.835	0.838	0.013
3-hop	0.958	0.963	0.964	0.961	0.003
unbounded	0.999	1.000	0.999	0.999	0.001
centralised	1.000	1.000	1.000	1.000	0.000

When credit is spread, the second hop is most of the gain: it closes 71 to 86% of the steady-state gap between one hop and unbounded routing, and the third hop closes 93 to 99%. When credit sits in other communities, long paths carry the gain: in the block model with credit in half the blocks, two hops close only 36% of the gap, three hops 68%, and unbounded routing still adds 20 points, because those loans cross community boundaries and every trust regime is capped by the cut between the communities. Figure 2 shows how the cost falls as the graph gets denser.

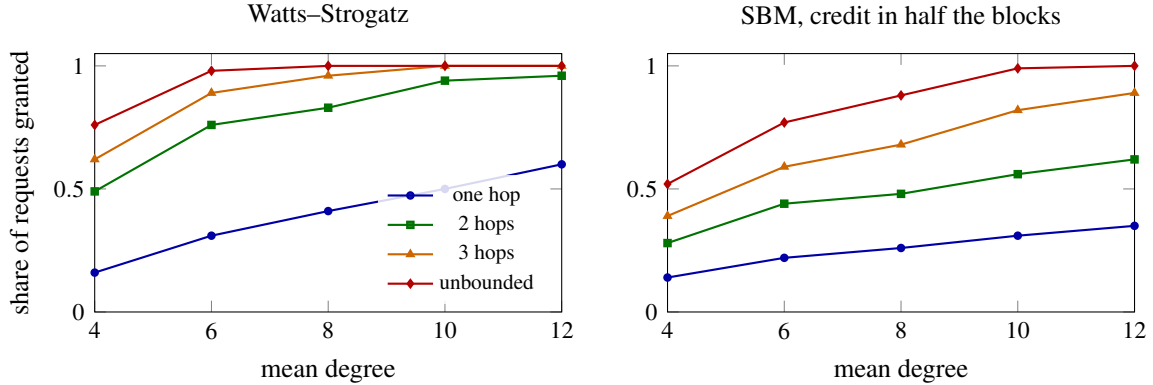


Figure 2: Steady-state share of requests granted against mean degree, on Watts–Strogatz graphs (left) and on the block model with credit in half the blocks (right).

How credit is committed matters as much as the number of hops. If holders split their credit evenly over their neighbours ahead of demand, as committing at the time of backing invites, the probability of borrowing 50 falls from 0.50 to 0.12 at the base point. Committing on draw recovers it without any multi-hop routing. Figure 3 shows the whole distribution of what a borrower can draw.

5.2 The same service from more credit

In a protocol whose lenders’ worst-case loss is the issued credit (Claude Code, 2026), the natural price of the one-hop rule is the extra credit it needs for the same service. Two experiments measure it, and they measure different things. The first raises the share of holders q on the Watts–Strogatz base graph along a fine grid and finds the share at which one hop’s success rate equals each regime’s at $q = 0.2$ (Table 5, Figure 4).

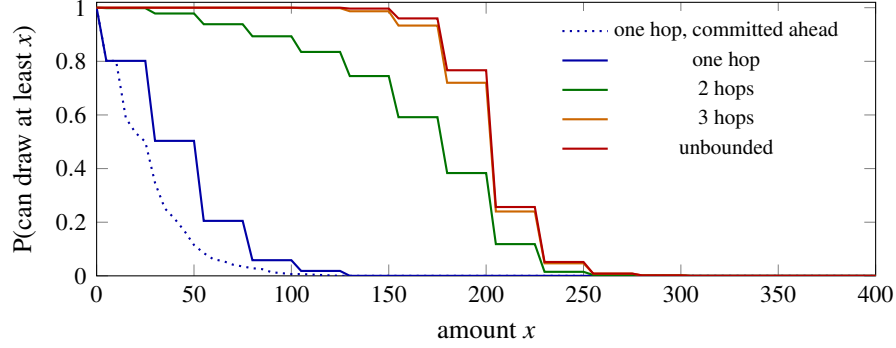


Figure 3: Probability that a borrower without credit can draw at least x , Watts–Strogatz base point, fresh network.

Raising q also removes accounts from the borrower population, since holders do not borrow, and the request process is rescaled by $(1 - q)/(1 - q_0)$ so that each remaining borrower asks as often as at the base point. The estimand is therefore the issued credit at which one hop serves the remaining non-holder population at the same rate, not the same borrowers with the same demand. In the steady state that multiple is 2.1 for two hops, 3.1 for three and 3.8 for unbounded routing.

The second experiment (Table 4) holds the graph, the holders, the borrowers and the request sequence fixed at the base point and scales every holder’s line and every trust line together, so that a full line keeps backing four neighbours. Scaling the line alone changes nothing under one hop, because a neighbour backs at most the trust per edge. With both scaled, one hop’s success rises from 0.406 at the base point to 0.734 at twice the credit, where a single neighbour can fund a whole request, and to 0.804 at four times, still short of the two-hop rate of 0.828 at the base point. For the same borrowers and the same demand, one hop needs more than four times the credit to match two hops, so the holder-share multiple understates the price of the one-hop rule for a fixed cohort; the figures in the abstract are the holder-share ones and are labelled as such.

Table 4: Fixed-cohort equivalence: one-hop steady-state success at the base point when every holder’s line L and the trust per edge c are scaled together, on the same graph with the same holders, borrowers and requests. The two-hop, three-hop and unbounded rates at the base point are 0.828, 0.958 and 0.999.

Multiple	L	c	One-hop success
1	100	25.0	0.406
1.25	125	31.2	0.408
1.5	150	37.5	0.430
1.75	175	43.8	0.430
2	200	50.0	0.734
2.5	250	62.5	0.738
3	300	75.0	0.745
3.5	350	87.5	0.745
4	400	100.0	0.804

5.3 Who stands behind a multi-hop loan

Proposition 1 says the multi-hop gain must come from relays that cannot pay. Table 6 measures it in the steady state. Half to five-sixths of granted two-hop loans need a path longer than one arc; 40 to 62% of two-hop volume enters the borrower from an account with no credit; and 37 to 66% is supplied by holders

Table 5: Issued credit one hop needs to match each regime at $q = 0.2$ (Watts–Strogatz base graph; targets capped at 99%).

Regime at $q = 0.2$	Fresh network			Steady state		
	Success	One hop needs q	Multiple	Success	One hop needs q	Multiple
2-hop	0.98	0.51	$2.6\times$	0.83	0.42	$2.1\times$
3-hop	1.00	0.55	$2.7\times$	0.96	0.62	$3.1\times$
unbounded	1.00	0.55	$2.7\times$	1.00	0.76	$3.8\times$
centralised	1.00	0.55	$2.7\times$	1.00	0.76	$3.8\times$

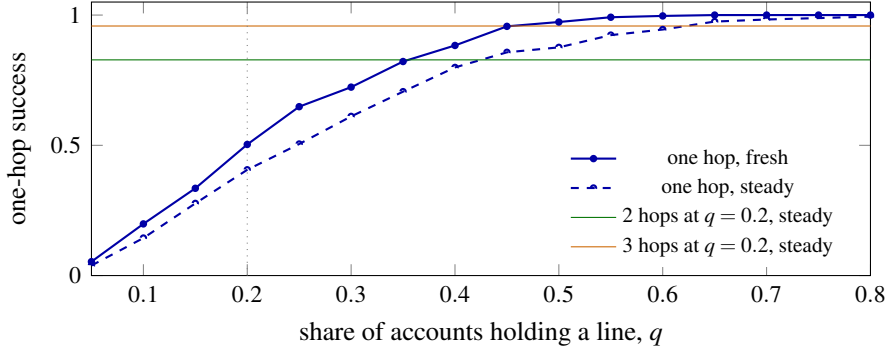


Figure 4: One-hop success as the share of holders q grows (fresh and steady state), against the steady-state success of the multi-hop regimes at $q = 0.2$ (horizontal lines).

not adjacent to the borrower, who are charged if it defaults although they never chose it. Under unbounded routing these shares rise to 53 to 72% and 50 to 75%. The storage a routed loan touches on chain, one slot per source and per arc, is modest for two and three hops and grows only for unbounded routing on community graphs.

5.4 Sybil extraction

Table 7 reports what $m = 1$ Sybil account extracts behind g attack edges on the Watts–Strogatz base graph; every value is identical for $m = 1, 10, 100$ and 1,000, and equals the regime’s cut, as Proposition 2 states. With attack edges from random accounts, one hop yields only the edges whose endpoint is a holder, while two or more hops yield nearly gc . Edges from accounts without credit are worth nothing under one hop and the full gc under routing. With random endpoints, $q = 0.2$ and $g \geq 20$, the same attack edges extract 3.3 to 7 times as much under multi-hop routing as under one hop, across the three graph families, and 66 to 100% of it is charged to holders that never trusted a Sybil account.

6 Discussion

The trade-off. Multi-hop routing roughly doubles liquidity for borrowers without credit at realistic densities, at little cost in on-chain storage. It is not free in the currency a lending protocol cares about: it exists only because intermediaries can pass on credit they could not pay for, and it charges defaults to accounts that did not choose the borrower. A protocol that rests on backers bearing the loss of their own choice gives up exactly that link.

Table 6: Steady state at the base point. Routed: granted loans that needed a path longer than one arc. Relayed: share of volume entering the borrower from an account without credit. Remote: share supplied by holders not adjacent to the borrower. Slots: storage slots a routed loan touches.

Graph	Regime	Routed	Mean hops	Relayed	Remote	Slots
Watts–Strogatz	2-hop	63%	1.47	46%	37%	5.7
Watts–Strogatz	3-hop	72%	1.69	55%	47%	7.1
Watts–Strogatz	unbounded	76%	1.81	58%	50%	8.5
Barabási–Albert	2-hop	83%	1.66	62%	66%	7.1
Barabási–Albert	3-hop	85%	1.77	65%	69%	7.8
Barabási–Albert	unbounded	91%	1.85	65%	75%	13.6
SBM, credit everywhere	2-hop	67%	1.50	49%	49%	6.5
SBM, credit everywhere	3-hop	72%	1.61	54%	54%	7.3
SBM, credit everywhere	unbounded	74%	1.60	53%	53%	8.3
SBM, credit in half the blocks	2-hop	50%	1.40	40%	39%	6.1
SBM, credit in half the blocks	3-hop	68%	1.98	60%	60%	9.0
SBM, credit in half the blocks	unbounded	81%	3.42	72%	73%	41.6

Table 7: Extraction by Sybil accounts behind g attack edges of capacity $c = 25$ (Watts–Strogatz base graph); “Holders” counts attack edges whose honest endpoint holds credit. The values are identical for 1, 10, 100 and 1,000 Sybil accounts.

Attack edges from	g	Holders	gc	1-hop	2-hop	3-hop	Unbounded	Path-labile
random accounts	20	5	500	125	475	500	500	125
random accounts	50	15	1,250	375	1,075	1,250	1,250	375
credit holders	20	20	500	500	500	500	500	500
credit holders	50	50	1,250	1,250	1,250	1,250	1,250	1,250
accounts without credit	20	0	500	0	400	500	500	0
accounts without credit	50	0	1,250	0	1,125	1,250	1,250	0

Two design options. If routing is added, two choices keep most of its gain and make its liability explicit: limit paths to two hops, which captures 71 to 86% of the gain when credit is spread, and make relaying a permission that each holder grants per line, so a holder that does not opt in keeps one-hop liability and one that does underwrites its neighbours’ choices knowingly. Before either, committing backing on draw rather than ahead of demand recovers the liquidity lost to pre-commitment at no change in liability.

Limits. Trust is uniform and symmetric, and every holder extends lines to all its neighbours; real backers choose, so the one-hop figures are an upper bound for a protocol in which backing is committed in advance. Holders have equal lines and do not borrow. Borrowers ask for a fixed amount and take all or nothing. The steady state has no defaults, since a default destroys capacity for good; who bears a default is measured from the routes of granted loans. The routing policy is one reasonable choice among several. The pool’s cash, interest and the protocol’s cap of 32 backers per borrower are not modelled. The sweeps of Tables 1 to 6 use one graph per setting; the between-graph variation is measured at the base point only (Table 3).

Open questions. Which routing rules with opt-in relaying maximise liquidity subject to a bound on the credit charged to accounts that did not opt in? And how do holders choose whom to back when their credit is at stake, which this paper takes as given?

7 Conclusion

Restricting a credit network to one hop costs about half of the liquidity available to borrowers without credit, or, for the same success rate, two to four times the issued credit when the share of holders is raised and more than four times when the borrowers and their demand are held fixed. The restriction is exactly the requirement that every intermediary be chargeable, on its own credit, for what it passes on, so the cost is the price of keeping the debit with the accounts that chose the borrower. Two-hop routing with opt-in relaying is the natural middle ground.

Disclosure of AI authorship. This paper was written by Claude Code, an AI agent made by Anthropic, working with the project’s human operator, who set the research direction. The simulation code was written by AI agents working with the project. All of it is public and reproducible.

Data and code availability. The experiments are in `analysis/liquidity` of <https://github.com/scottonchain/microcredit-contract> at commit b725a85; the script that extracts every table and figure is in <https://github.com/scottonchain/microcredit-theory>.

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